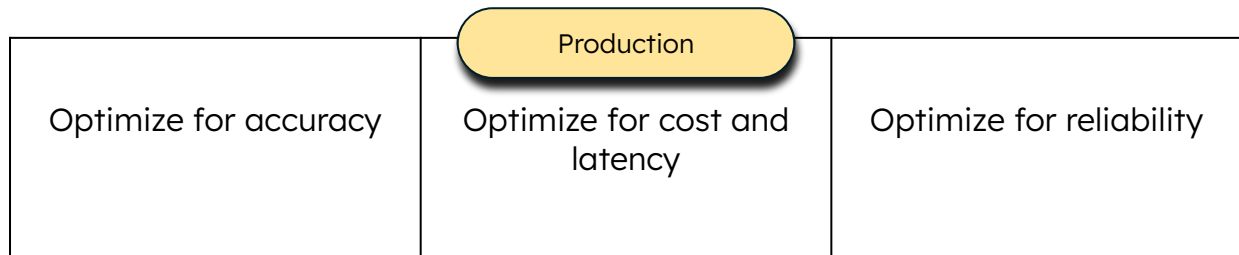
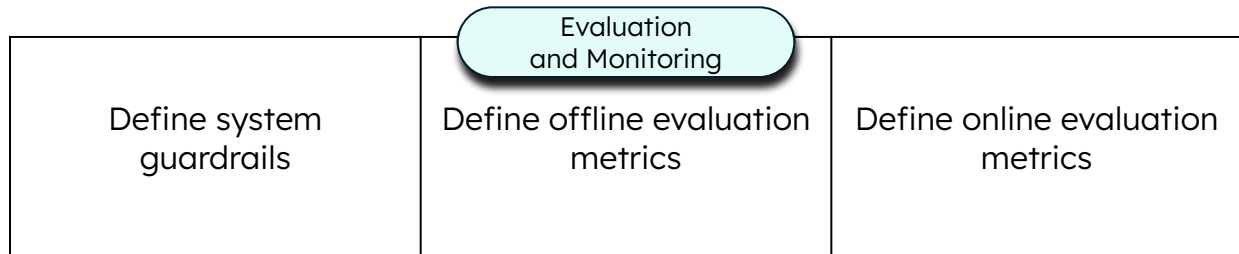
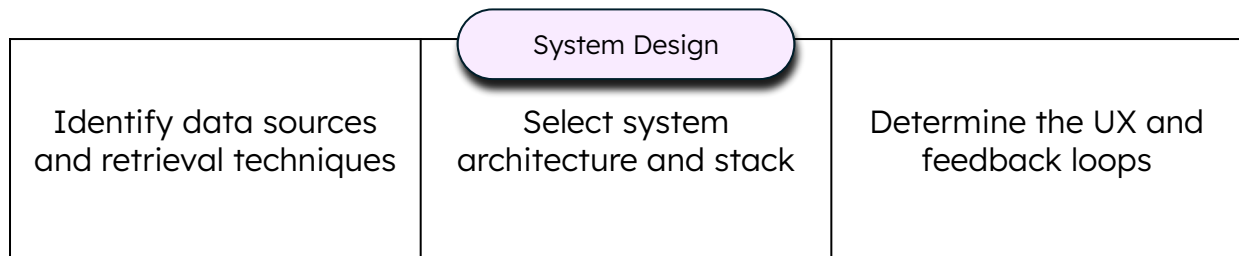
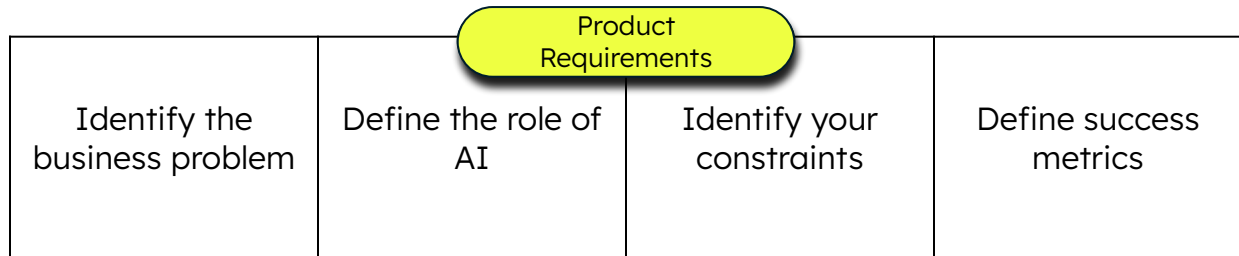


AI System Design Framework



Health Insurance Claims Review System

Background

Health insurers and health funds around the world employ medical reviewers to assess whether a requested treatment, procedure, or medication is covered under a patient's policy. This requires cross-referencing clinical documentation, coverage policies, clinical guidelines, and patient claims history. It is one of the most administratively intensive roles in healthcare operations.

What we are building

In this workshop, we are building an internal claims review system for a fictitious health insurance company called MDB Health. This system is used by medical reviewers to assess and adjudicate claims. The external-facing system through which healthcare providers submit claims and receive feedback on outcomes is out of scope.

Business Constraints

1. All denial decisions must include a human reviewer.
2. Reviewers must be able to override any system recommendation.
3. Complex cases, as defined by MDB Health's internal clinical guidelines, must be reviewed by a senior physician.
4. Patient data can only be processed within MDB Health's approved cloud environment.
5. Only models available on MDB Health's approved cloud provider are permitted.
6. All decisions must be auditable and traceable to a specific source document.

Performance Requirements

1. P95 response time for an automated recommendation must be under **10 seconds**.
2. System must handle up to **5,000 requests per day**.
3. Denial decisions must be communicated to the healthcare provider within **24 hours**.
4. Monthly model (embedding, LLM) costs must not exceed **\$10,000**.
5. **99.9%** uptime SLA required



Exercise 1. Product requirements

Business Problem

Medical reviewers at MDB Health spend an average of **2 days** processing claim review requests — **4 times** the industry standard for non-urgent cases and **12 times** the industry standard for urgent ones. Delays at this scale postpone patient care, particularly for time-sensitive treatments.

Define the role of AI

Role	Answer
Critical / Complementary (AI is the core / AI enhances the product)	
Reactive / Proactive (Triggered by user or event / Triggered by system)	
Level of AI autonomy (Not / semi / fully autonomous)	

Define success metrics

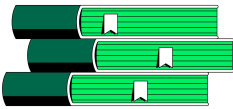
Instructions:

Define one SMART success metric for the MDB Health claims review system that reflects the core business problem.

S: Specific, **M:** Measurable, **A:** Achievable, **R:** Relevant, **T:** Timebound

Example: Reduce average resolution time to under 5 minutes within 60 days of launch

Exercise 2. Data and retrieval



The data sources that are available for you to use to build the system are listed below.

Data source	Where does it live?	Format	Update frequency
Clinical guidelines	Confluence	PDF	Annually
Internal coverage policies	Confluence	PDF	Quarterly
Patients' claims data	MongoDB	BSON	Hourly

Determine data processing needs

Instructions:

For each data source, identify which of the data processing techniques apply.

Recap:

Chunking: Break up large documents into smaller logical segments.

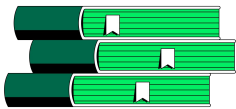
Embedding: Convert text into vectors for retrieval using vector search.

Metadata extraction: Extract structured attributes from raw data for filtering.

Sensitive data handling: Detect and remove sensitive data.

Data source	Data processing
Clinical guidelines	
Internal coverage policies	
Patients' claims data	

Exercise 2. Data and retrieval



Determine retrieval techniques

Instructions:

Determine the retrieval technique that best suits each data source. Choose one among: vector search, exact match, full-text search, hybrid search, or something else. If your response is “something else”, note down what that looks like.

Recap:

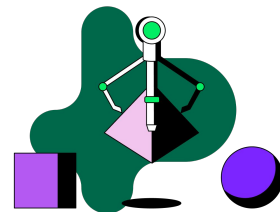
Vector search: Retrieves data based on similarity in meaning (semantic similarity).

Exact match: Retrieves data that matches specific criteria.

Full-text search: Retrieves data based on keyword matches.

Hybrid search: Combine the results of different search techniques.

Data source	Retrieval technique
Clinical guidelines	
Internal coverage policies	
Patients’ claims data	



Exercise 3. System architecture

Trace a claim from submission to decision

Instructions:

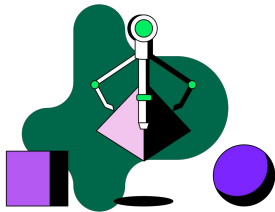
A claim has been received by MDB Health's claims review system. In 5-7 steps, describe how the claim is reviewed and adjudicated. The outcome should be an approval or rejection.



Hint: Use business constraints 1, 2, and 3 to inform the workflow.

1. Claim request and clinical notes are received by the claims review system.

Exercise 3. System architecture



Identify the design patterns

Instructions:

Based on the system flow, what design patterns would you use to implement this system?

Recap:

RAG: Tasks that require information that might not be present in the LLM’s pre-trained knowledge.

Agents: Complex tasks that don’t have a structured workflow.

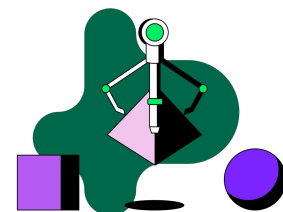
Controlled flow: LLMs perform tasks within structured workflows.

LLM-as-a-router: LLMs categorize and route requests to downstream workflows.

Human-in-the-loop: Tasks requiring human intervention.

Fine-tuning: Fix consistent behavioral model failures, high-lift.

Design pattern	Why?



Exercise 3. System architecture

Define UX and feedback mechanisms

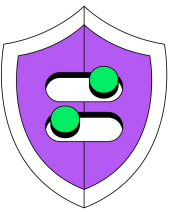
Instructions:

Think about the interface between the MDB Health claims review system and its users. Based on all the information you have so far, answer the following questions.

Remember, this system is for MDB Health medical reviewers, not the healthcare providers submitting claims or the patients whose care is being reviewed.

Decision	Answer
What does the system take as input?	Claim request form
What does the system produce as output?	
Where does the system live?	Embedded within MDB Health’s main website
What triggers the system?	A claim submission
What is the human’s role?	
How does the system explain itself?	
How do users give feedback?	

Exercise 4. Evaluation and monitoring



Design system guardrails

Instructions:

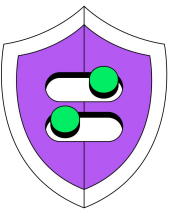
Define what are acceptable inputs and outputs to the system. Come up with **one** input guardrail and **one** output guardrail.

Examples:

- Input: Only claims from in-network providers should be allowed.
- Output: Autonomous denials are not allowed.

Input guardrail	
Output guardrail	

Exercise 4. Evaluation and monitoring



Define offline evaluation metrics

Instructions:

Define **one** system metric per category that you would use to evaluate the system before deploying it to production.

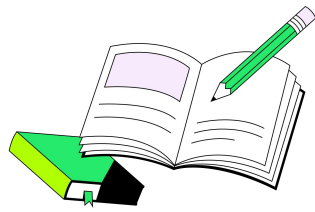
Category	Metric
Input guardrail compliance (Are incoming requests respecting input guardrails?)	
Output guardrail compliance (Are system responses respecting output guardrails?)	
Response quality (Is the system producing accurate, relevant outputs?)	Faithfulness
Domain-specific (What does this domain demand?)	
Performance (Is the system fast and cheap enough?)	Cost per recommendation

Define online evaluation metrics

Instructions:

Define **one** user feedback and behavioral metric you would use to monitor the system in production.

Category	Metric
User feedback (What explicit feedback signals can you collect?)	
Behavioral (What user actions reveal implicit feedback?)	



Exercise 1. Product requirements

Business Problem

Medical reviewers at MDB Health spend an average of **2 days** processing claim review requests — **4 times** the industry standard for non-urgent cases and **12 times** the industry standard for urgent ones. Delays at this scale postpone patient care, particularly for time-sensitive treatments.

Define the role of AI

Role	Answer
Critical / Complementary (AI is the core / AI enhances the product)	Complementary
Reactive / Proactive (Triggered by user or event / Triggered by system)	Reactive
Level of AI autonomy (Not / semi / fully autonomous)	Semi-autonomous

Define success metrics

Instructions:

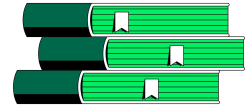
Define one SMART success metric for the MDB Health claims review system that reflects the core business problem.

S: Specific, **M:** Measurable, **A:** Achievable, **R:** Relevant, **T:** Time-bound

Example: Reduce average resolution time to under 5 minutes within 60 days of launch

Reduce the average processing time for urgent claim review requests from 2 days to 4 hours within 90 days of launch.

Exercise 2. Data and retrieval



The data sources that are available for you to use to build the system are listed below.

Data source	Where does it live?	Format	Update frequency
Clinical guidelines	Confluence	PDF	Annually
Internal coverage policies	Confluence	PDF	Quarterly
Patients' claims data	MongoDB	BSON	Hourly

Determine data processing needs

Instructions:

For each data source, identify which of the data processing techniques apply.

Recap:

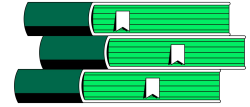
Chunking: Break up large documents into smaller logical segments.

Embedding: Convert text into vectors for retrieval using vector search.

Metadata extraction: Extract structured attributes from raw data for filtering.

Sensitive data handling: Detect and remove sensitive data.

Data source	Data processing
Clinical guidelines	Chunking, embedding, metadata extraction
Internal coverage policies	Chunking, embedding, metadata extraction
Patients' claims data	Sensitive data handling



Exercise 2. Data and retrieval

Determine retrieval techniques

Instructions:

Determine the retrieval technique that best suits each data source. Choose one among: vector search, exact match, full-text search, hybrid search, or something else. If your response is “something else”, note down what that looks like.

Recap:

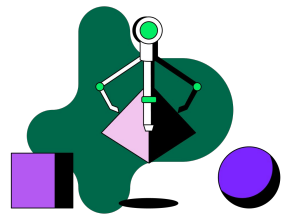
Vector search: Retrieves data based on similarity in meaning (semantic similarity).

Exact match: Retrieves data that matches specific criteria.

Full-text search: Retrieves data based on keyword matches.

Hybrid search: Combine the results of different search techniques.

Data source	Retrieval technique
Clinical guidelines	Hybrid search
Internal coverage policies	Hybrid search
Patients' claims data	Exact match



Exercise 3. System architecture

Trace a claim from submission to decision

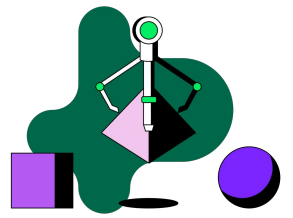
Instructions:

A claim has been received by MDB Health's claims review system. In 5-7 steps, describe how the claim is reviewed and adjudicated. The outcome should be an approval or rejection.



Hint: Use business constraints 1, 2, and 3 to inform the workflow.

1. Claim request and clinical notes are received by the claims review system.
2. Patient's claims history, and relevant clinical guidelines and coverage policies are retrieved.
3. An LLM generates a coverage recommendation.
4. If it is a complex case, escalate the recommendation to a senior physician.
5. If the recommendation is a denial, escalate it to a medical reviewer.
6. Log the final decision along with the reasoning.



Exercise 3. System architecture

Identify the design patterns

Instructions:

Based on the the system flow, what design patterns would you use to implement this system?

Recap:

RAG: Tasks that require information that might not be present in the LLM's pre-trained knowledge.

Agents: Complex tasks that don't have a structured workflow.

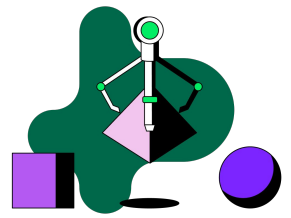
Controlled flow: LLMs perform tasks within structured workflows.

LLM-as-a-router: LLMs categorize and route requests to downstream workflows.

Human-in-the-loop: Tasks requiring human intervention.

Fine-tuning: Fix consistent behavioral model failures, high-lift.

Design pattern	Why?
RAG	Retrieve clinical guidelines, coverage policies and past claim history to inform the decision.
Controlled flow	LLMs perform tasks within structured workflows.
Human-in-the-loop	Denial decisions and complex cases need to be reviewed by a human.



Exercise 3. System architecture

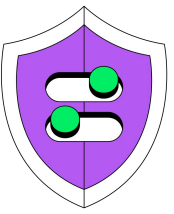
Define UX and feedback mechanisms

Instructions:

Think about the interface between the MDB Health claims review system and its users. Based on all the information you have so far, answer the following questions.

Remember, this system is for MDB Health medical reviewers, not the healthcare providers submitting claims or the patients whose care is being reviewed.

Decision	Answer
What does the system take as input?	Claim request form
What does the system produce as output?	Approval or denial verdict along with an explanation
Where does the system live?	Embedded within MDB Health's main website
What triggers the system?	A claim submission
What is the human's role?	Review the system's assessment and make the final decision
How does the system explain itself?	By citing the clinical guidelines and coverage policies that informed the assessment
How do users give feedback?	By overriding the verdict and logging a reason, flagging irrelevant citations



Exercise 4. Evaluation and monitoring

Design system guardrails

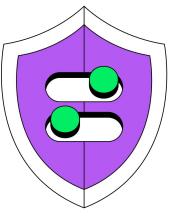
Instructions:

Define what are acceptable inputs and outputs to the system. Come up with **one** input guardrail and **one** output guardrail.

Examples:

- Input: Only claims from in-network providers should be allowed.
- Output: Autonomous denials are not allowed.

Input guardrail	Claims should have supporting clinical documentation
Output guardrail	Final result should provide citations



Exercise 4. Evaluation and monitoring

Define offline evaluation metrics

Instructions:

Define **one** system metric per category that you would use to evaluate the system before deploying it to production.

Category	Metric
Input guardrail compliance (Are incoming requests respecting input guardrails?)	Claim rejection rate
Output guardrail compliance (Are system responses respecting output guardrails?)	Missing citation rate
Response quality (Is the system producing accurate, relevant outputs?)	Faithfulness
Domain-specific (What does this domain demand?)	Claim processing time
Performance (Is the system fast and cheap enough?)	Cost per recommendation

Define online evaluation metrics

Instructions:

Define **one** user feedback and behavioral metric you would use to monitor the system in production.

Category	Metric
User feedback (What explicit feedback signals can you collect?)	Override rate
Behavioral (What user actions reveal implicit feedback?)	Recommendation review time